

Physics 4220H: Assignment I

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Problem 1. Derive the four typical boundary conditions given below from (the complete) Maxwell's equations. Please include diagrams as needed (drawn by hand is fine):

$$\begin{aligned}\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp &= \sigma_f \\ \mathbf{E}_1^\parallel - \mathbf{E}_2^\parallel &= \mathbf{0} \\ \frac{1}{\mu_1} \mathbf{B}_1^\parallel - \frac{1}{\mu_2} \mathbf{B}_2^\parallel &= \mathbf{K}_f \times \hat{\mathbf{n}} \\ B_1^\perp - B_2^\perp &= 0\end{aligned}$$

Solution 1. Starting with Maxwell's equations in linear media,

$$\epsilon \oint_S \mathbf{E} \cdot d\mathbf{a} = Q_{f_{enc}} \quad (1)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (2)$$

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a} \quad (3)$$

$$\frac{1}{\mu} \oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = I_{enc} - \epsilon \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{a} \quad (4)$$

we construct two scenarios, one of a charge-carrying (free charge density σ_f) surface, the other of a sheet of surface current density $\mathbf{K}_f$. We take region 1 to be the above region, and 2 the below. I've attached handmade drawings when I would usually do something in TikZ as I can't bring myself to put in the effort needed when there are very good drawings already in Griffiths.

- (a) If we take the charge-carrying surface and place a “Gaussian pillbox” so that it is cut in half by the surface, then take the pillbox to be infinitesimally thin, and the sheet to be very large, the electric field produced by the sheet is entirely normal to said sheet. This allows us to eliminate everything except the top and bottom (parallel to the sheet) components of the integral in eq. (1) which is run over the pillbox. Since these two pieces of the integral experience a uniform electric field (we force σ_f to be such that this is so),

$$\epsilon_1 \mathbf{E}_1 \cdot \mathbf{A}_1 + \epsilon_2 \mathbf{E}_2 \cdot \mathbf{A}_2 = \sigma_f A.$$

$\mathbf{A}_1$ and $\mathbf{A}_2$ have the same magnitude, A , but one is pointing in line with $\mathbf{E}_1$ and the other in line but opposite in direction to $\mathbf{E}_2$, giving us the negative sign we want. The perpendicular components are picked out by the dot product with the area-normal vectors, giving

$$\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp = \sigma_f.$$

- (b) Now, still considering the sheet of charge, we replace the pillbox with a closed loop $\mathcal{C}$, and shrink its “height” (the sides perpendicular to the sheet) to again be infinitesimal, the left-hand side of eq. (3) drops to zero as the area of the surface shrinks to zero and the right-hand side can be cut down to just the contribution of the sides of $\mathcal{C}$ parallel to the sheet,

$$\mathbf{E}_1 \cdot \boldsymbol{\ell} - \mathbf{E}_2 \cdot \boldsymbol{\ell} = 0$$

where we picked up the negative sign due to the decision to traverse the curve in a right-handed manner. If we divide out the magnitude of $\boldsymbol{\ell}$, leaving us with the components of $\mathbf{E}_1$ and $\mathbf{E}_2$ along $\boldsymbol{\ell}$ we obtain the desired result,

$$\mathbf{E}_1^\parallel - \mathbf{E}_2^\parallel = \mathbf{0}.$$

- (c) Now considering the sheet of surface current $\mathbf{K}_f$ and a loop placed so that the current is perpendicular to the area-normal of the surface bounded by the loop, the left-hand side surface integral vanishes as the surface does (because we shrink the height of the loop), and the right-hand side integral picks up only the contribution due to the current-sheet-parallel components of $\mathbf{B}$,

$$(\mathbf{B}_1 - \mathbf{B}_2) \cdot \boldsymbol{\ell} = I_{f_{enc}}.$$

Since the loop is too thin to contain any volume current, the only contribution can be due to our constructed free surface current $\mathbf{K}_f$, the component of which that passes “through” the loop (though it is infinitely thin, so is the surface current) is given by

$$I_{f_{enc}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \boldsymbol{\ell})$$

where $\mathbf{n}$ is the current-sheet surface normal vector. Using the scalar triple product this transforms to

$$I_{f_{enc}} = (\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \boldsymbol{\ell}$$

which allows us to knock off the shared dotting with $\boldsymbol{\ell}$ and obtain the desired result,

$$\mathbf{B}_1^\parallel - \mathbf{B}_2^\parallel = \mathbf{K}_f \times \hat{\mathbf{n}}.$$

- (d) Here we apply the exact same reasoning as in (a), just that the Maxwell equation we use, eq. (2), is zero on the LHS.

Problem 2. Derive the charge continuity equation,

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\boldsymbol{\nabla} \cdot \mathbf{J}(\mathbf{r}, t)$$

from (the complete) Maxwell’s Equations.

Solution 2. Ampère’s law states that

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Taking the divergence of both sides of this we get that

$$\boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) = 0 = \mu_0 \boldsymbol{\nabla} \cdot \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \boldsymbol{\nabla} \cdot \mathbf{E}$$

which can be rearranged to

$$\boldsymbol{\nabla} \cdot \mathbf{J} = -\epsilon_0 \frac{\partial}{\partial t} \boldsymbol{\nabla} \cdot \mathbf{E}. \quad (5)$$

Gauss’ law then states that

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (6)$$

Plugging eq. (6) into eq. (5) we get the continuity equation,

$$\boldsymbol{\nabla} \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

Problem 3.

- (a) Show that, in linear media, the Poynting vector can be given by $\mathbf{S} = \mathbf{E} \times \mathbf{H}$, and that the energy density is $u_{EM} = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$.
- (b) Describe in words the conceptual difference between the Poynting vector and the total electromagnetic energy density. Can you give an example of when computing one might be more relevant than the other?

Solution 3.

(a) I'll follow basically the same process we did in class except using Maxwell's Equations in linear media,

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho_f \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}.\end{aligned}$$

Now the work done by an electromagnetic field on a charge in infinitesimal timestep dt (over which a length $d\ell$ is traveled) is $dW = \mathbf{F} \cdot d\ell = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\ell$. Noting that $\mathbf{v} = \frac{d\ell}{dt}$ we can rewrite this as

$$dW = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt.$$

Noting then that $\mathbf{v} \times \mathbf{B}$ by definition of the cross product gives a vector perpendicular to $\mathbf{v}$ and $\mathbf{B}$, dotting the output with $\mathbf{v}$ will give zero, hence, dividing through by dt ,

$$\frac{dW}{dt} = \mathbf{E} \cdot q\mathbf{v}.$$

Then since $q = \int_{\mathcal{V}} \rho(\mathbf{r}') dr'$,

$$\frac{dW}{dt} = \mathbf{E} \cdot \int_{\mathcal{V}} \rho(\mathbf{r}') \mathbf{v} dr' = \int_{\mathcal{V}} \mathbf{E} \cdot \mathbf{J} dr'$$

provided $\mathbf{E}$ is position independent in $\mathcal{V}$ and since $\rho\mathbf{v}$ is a charge moving, i.e. a current. Now applying Ampère's law in media,

$$\begin{aligned}\mathbf{J} &= \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}, \\ \frac{dW}{dt} &= \int_{\mathcal{V}} \mathbf{E} \cdot (\nabla \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} dr' .\end{aligned}$$

Now to fiddle with the first term we use the vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = (\nabla \times \mathbf{E}) \cdot \mathbf{H} - \mathbf{E} \cdot (\nabla \times \mathbf{H})$ and obtain

$$\frac{dW}{dt} = \int_{\mathcal{V}} (\nabla \times \mathbf{E}) \cdot \mathbf{H} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} dr'.$$

Now applying Faraday's law to the first term,

$$\frac{dW}{dt} = - \int_{\mathcal{V}} \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \nabla \cdot (\mathbf{E} \times \mathbf{H}) dr'.$$

Now using the bit of a priori knowledge from class regarding the chain-rule differentiation trick we can look more closely at the first two terms. Since $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu\mathbf{H}$ where $\mu = \mu_0(1 + \chi_m)$ and

$$\frac{1}{\mu} \frac{\partial |\mathbf{B}|^2}{\partial t} = \frac{2}{\mu} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = 2\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}$$

we can rewrite the first term as

$$\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial \mathbf{B} \cdot \mathbf{H}}{\partial t}$$

and the second term using the same chain rule approach with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_0(1 + \chi_e)\mathbf{E}$ as

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{1}{2} \frac{\partial \mathbf{E} \cdot \mathbf{D}}{\partial t}.$$

The main expression then becomes

$$\frac{dW}{dt} = - \int_{\mathcal{V}} \frac{1}{2} \frac{\partial}{\partial t} [\mathbf{B} \cdot \mathbf{H} + \mathbf{E} \cdot \mathbf{D}] dr' - \int_{\mathcal{V}} \nabla \cdot (\mathbf{E} \times \mathbf{H}) dr'.$$

We can fiddle with the last term with the divergence theorem (and pull out the time derivative in the first) to obtain

$$\frac{dW}{dt} = - \frac{1}{2} \frac{d}{dt} \int_{\mathcal{V}} [\mathbf{B} \cdot \mathbf{H} + \mathbf{E} \cdot \mathbf{D}] dr' - \oint_{\mathcal{S}} (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{a}.$$

I think at this point we can just look at Poynting's theorem in vacuum and compare the two. Performing such a comparison makes it immediately obvious that the term in the first integral is an analogue to u_{EM} , the electromagnetic energy density and the term in the second integral is the Poynting vector.

- (b) To my understanding, due to the way the dW/dt equation is written, the u_{EM} term represents the total energy stored in the fields in a volume $\mathcal{V}$. The Poynting vector integral term represents the flow of energy through $\mathcal{S}$, the surface which bounds $\mathcal{V}$. I can't come up with a specific situation, but it would be more useful to have the Poynting vector when you want to know where the energy is going, but (maybe) it would be easier (I kind of doubt it) to find energy density w.r.t time if you just want to know how energy is changing in a region.

Problem 4. The Dirac Quantization Condition is a famous rule in quantum mechanics that states that if even a single magnetic monopole exists in the universe, electric charge must be quantized. There is a famously cute (if technically improper) way to “show” this, involving quantization of field angular momentum:

Suppose that you had an electric point charge, q_e , separated by distance R from a magnetic point charge (or monopole), q_m . Put the monopole at the origin, and the electric charge at $\mathbf{R}$. The magnetic and electric fields emanating from these charges would be

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q_m}{r^3} \mathbf{r}$$

and

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_e}{|\mathbf{r} - \mathbf{R}|^3} (\mathbf{r} - \mathbf{R}).$$

- (a) Calculate the field momentum density, $\mathbf{p}_{EM}$, of this arrangement (known as a Thompson Dipole, btw), as a function of r , and comment on the direction of this momentum density.
- (b) From this, find the “field angular momentum” possessed by this arrangement, and comment on
- the meaning (or strangeness) of the functional dependence of this angular momentum on the distance between the two “charges”,
 - the direction of this angular momentum,
 - whether it is intuitive to you that static “charges” can set up fields arranged with nonzero angular momentum.

[Note, the integral involved is notoriously difficult, usually requiring several shortcuts or tricks. If needed, you may invoke the following result:

$$\int \frac{R}{r} \frac{1}{|\mathbf{r} - \mathbf{R}|} \left[\hat{\mathbf{R}} - \hat{\mathbf{r}} (\hat{\mathbf{r}} \cdot \hat{\mathbf{R}}) \right] dV = 4\pi \hat{\mathbf{R}}$$

- (c) A fundamental requirement of quantum mechanics is that angular momentum be quantized in units of $\hbar/2$. Use this fact, together with your result from (b) to argue for Dirac quantization.

Solution 4. (a) Directly from the given fields (I'm using $\mathbf{g}$ because that's what the book uses to make momentum density distinct from momentum),

$$\mathbf{g}_{EM} = \epsilon_0 \mathbf{E} \times \mathbf{B} = \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} (\mathbf{r} - \mathbf{R}) \times \mathbf{r}$$

and since $(\mathbf{r} - \mathbf{R}) \times \mathbf{r} = (-\mathbf{R}) \times \mathbf{r} = -|\mathbf{r}||\mathbf{R}| \sin(\theta) \hat{\mathbf{z}}$, which gives a new vector pointing into the page. Not sure if I can calculate the magnitude of this vector beyond the product of magnitudes and the angle between $\mathbf{r}$ and $\mathbf{R}$ as given above. To be explicit,

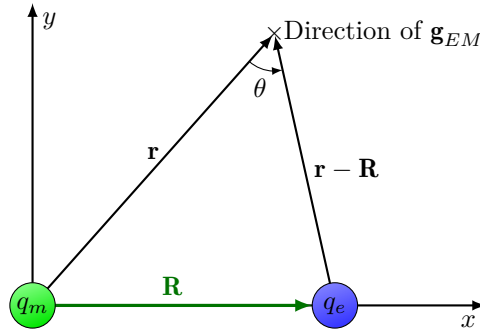
$$\mathbf{g}_{EM} = -\frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} |\mathbf{r}||\mathbf{r} - \mathbf{R}| \sin(\theta) \hat{\mathbf{z}} = -\frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^2 |\mathbf{r} - \mathbf{R}|^2} \sin(\theta) \hat{\mathbf{z}},$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{r} - \mathbf{R}$.

- (b) I find the book a little confusing here, it defines $\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g}$ as angular momentum, but had previously called $\mathbf{g}$ the momentum *density*, so is $\boldsymbol{\ell}$ the *angular* momentum density? I think so? That is the assumption I will proceed under, as this question mentions an integral which only makes sense to me in the context of a density. The angular momentum density for us is

$$\boldsymbol{\ell} = \mathbf{r} \times \mathbf{g}_{EM} = -\frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^2 |\mathbf{r} - \mathbf{R}|^2} \sin(\theta) \mathbf{r} \times \hat{\mathbf{z}} = -\frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r |\mathbf{r} - \mathbf{R}|^2} \sin(\theta) \hat{\mathbf{r}} \times \hat{\mathbf{z}}.$$

So we care about the result of $\mathbf{r} \times \hat{\mathbf{z}}$. At this point I feel the need to draw a picture, in the below image $\hat{\mathbf{z}}$ is out of the page.



I think we end up with

$$|\ell| = -\frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r |\mathbf{r} - \mathbf{R}|^2} \sin(\theta) \sin(\phi)$$

where ϕ is the (I want to say polar) angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$. I am having real trouble conceptualizing the direction of this, it would depend on where $\mathbf{r}$ is obviously. I am having less trouble conceptualizing that regardless of the direction, it's symmetric about the axis of the dipole, so only the components along $\mathbf{R}$ will survive the integration to get the angular momentum from the angular momentum density. So I'll just say then that the angular momentum will be along $\mathbf{R}$ and sweep under the rug that I can't figure out where the angular momentum density is pointing, or rather a function that tells you where it's pointing given $\hat{\mathbf{r}}$. After a bit of thought and trying to work backwards from the given integral formula, I realized that said formula looks like the remains of a vector triple product. Looking back at the original expression,

$$\begin{aligned} \ell &= \epsilon_0 \mathbf{r} \times (\mathbf{E} \times \mathbf{B}) \\ &= \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} \mathbf{r} \times [(\mathbf{r} - \mathbf{R}) \times \mathbf{r}] \\ &= \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} [(\mathbf{r} \cdot \mathbf{r})(\mathbf{r} - \mathbf{R}) - (\mathbf{r} \cdot (\mathbf{r} - \mathbf{R}))\mathbf{r}] \\ &= \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} [r^2 \mathbf{r} - r^2 \mathbf{R} - r^2 \mathbf{r} + (\mathbf{r} \cdot \mathbf{R})\mathbf{r}] \\ &= \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} [-r^2 \mathbf{R} + (\mathbf{r} \cdot \mathbf{R})\mathbf{r}] \\ &= \frac{\mu_0}{(4\pi)^2} \frac{q_m q_e}{r^3 |\mathbf{r} - \mathbf{R}|^3} \cancel{r} \cancel{R} [-\hat{\mathbf{R}} + (\hat{\mathbf{r}} \cdot \hat{\mathbf{R}})\hat{\mathbf{r}}] \\ &= -\frac{\mu_0}{(4\pi)^2} q_m q_e \frac{R}{r} \frac{1}{|\mathbf{r} - \mathbf{R}|^3} [\hat{\mathbf{R}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{R}})\hat{\mathbf{r}}]. \end{aligned}$$

So integrating the above expression using the given formula we obtain the angular momentum,

$$\mathbf{L} = -\frac{\mu_0}{4\pi} q_m q_e \hat{\mathbf{R}}.$$

To answer the sub questions, evidently the angular momentum points from the electric to the magnetic charge, the angular momentum has no dependence on the magnitude of $\mathbf{R}$, meaning these charges can be as far away as you can get them and their dipole has the same angular momentum, no it is not intuitive that these static charges can create angular momentum, and it feels like something similar to intrinsic angular momentum due to spin.

- (c) If angular momentum is to be quantized, then $|L| = n\hbar/2 \implies \frac{\mu_0}{4\pi} q_m q_e = n\hbar/2$ for $n \in \mathbb{Z}$. If this is true, then $\mu_0 q_m q_e = nh$. Since h and μ_0 are constants we can say that, in whatever units q_e and q_m are written, they must be integers themselves. Since $\mathbf{L}$ does not depend on distance, a single magnetic monopole would form this dipole structure with every charge everywhere, leading to the above condition on the integer nature of q_e to be enforced on every charge.